



Age- and Organ-Specific Lipid Droplet Remodeling under Rapamycin Treatment Revealed by Raman Imaging in Drosophila

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Background

mTOR Signaling Pathway

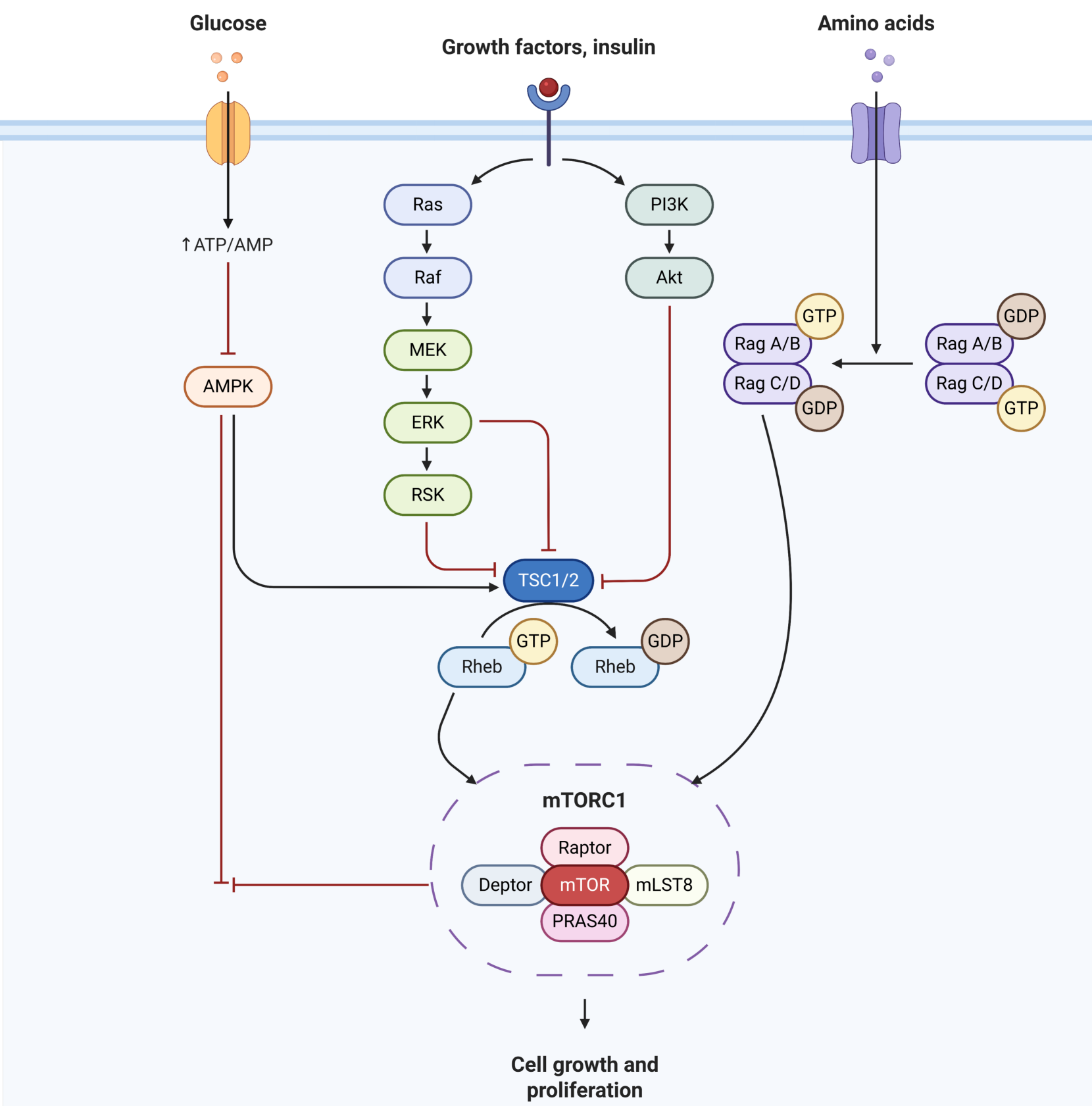


Figure 1. Signaling pathway of the Mechanistic Target of Rapamycin Complex 1 (mTORC1). Rapamycin acutely blocks this complex and inhibit downstream protein (s6k pathway) and lipid (SREBP pathway) synthesis.

Objectives

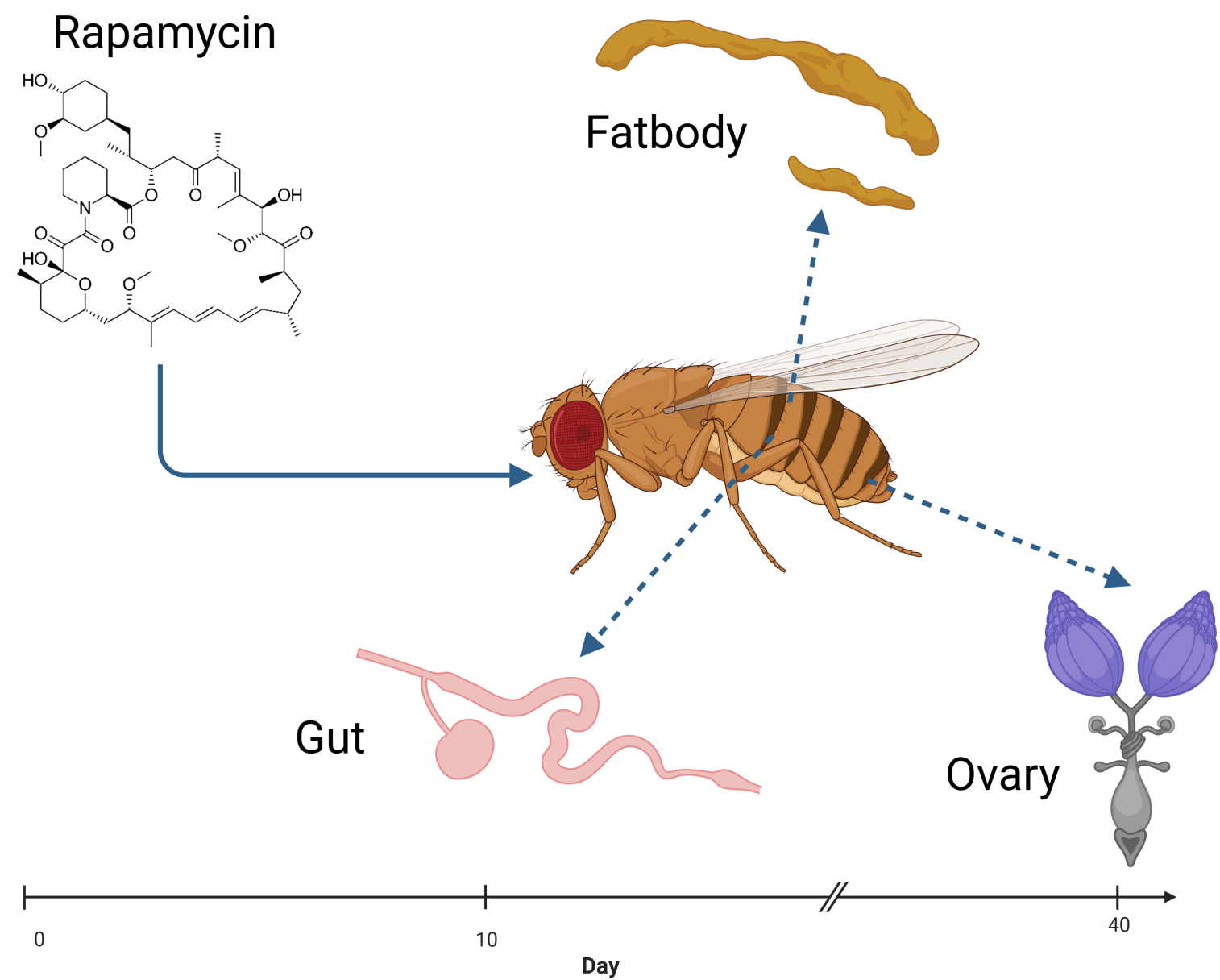


Figure 2. Rapamycin-mediated inhibition of mTORC1 extends lifespan in Drosophila, but its **organ-specific metabolic effects** remain poorly understood.

RESULTS

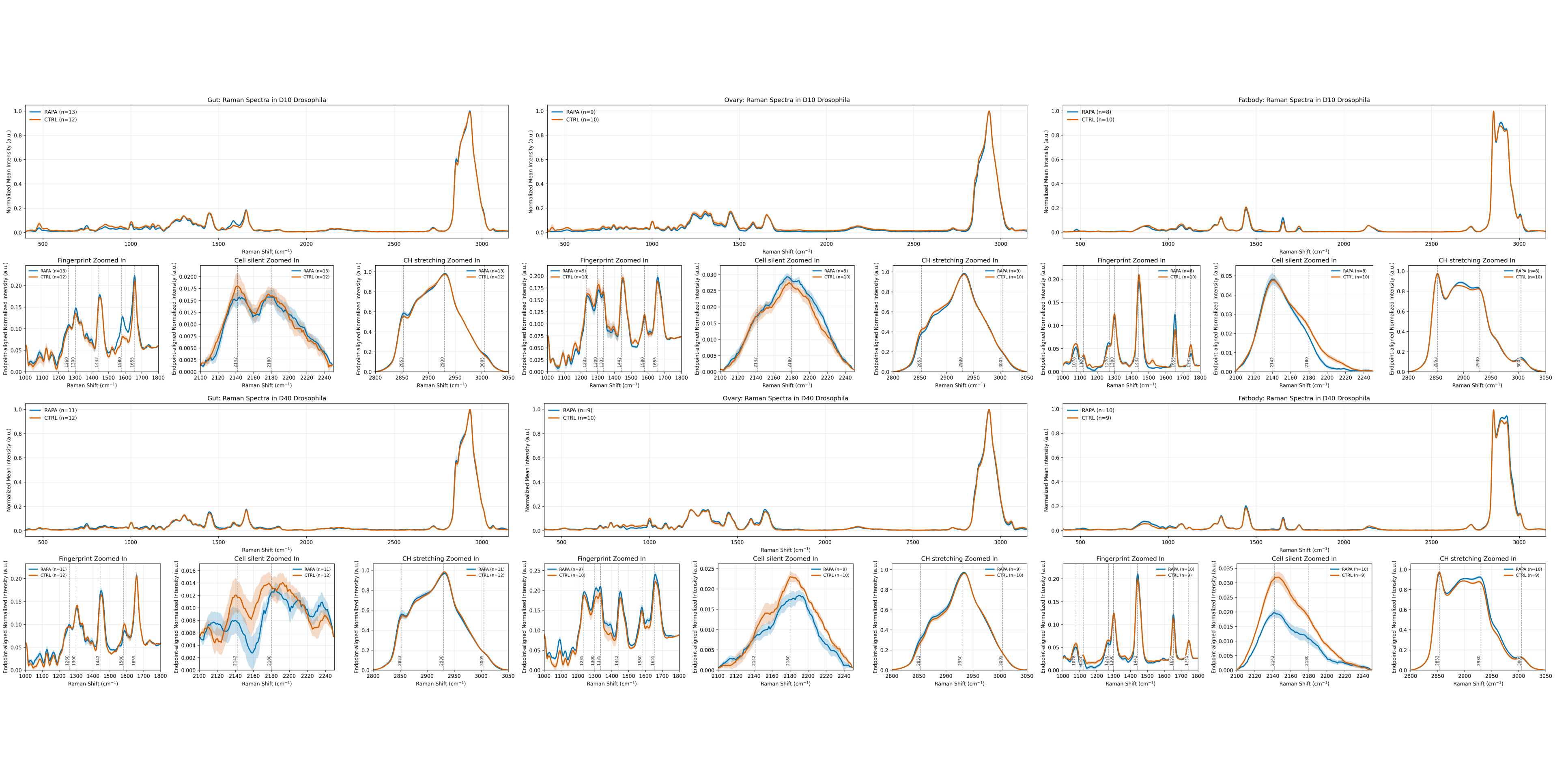


Figure 4. Raman spectral shape comparison between organs and age groups. The spontaneous Raman spectra is plotted from 400 cm⁻¹ to 3200 cm⁻¹, covering the fingerprint region, cell silent region, and CH stretching region with zoomed in peak comparisons showing the differences in peak shift and peak shape.

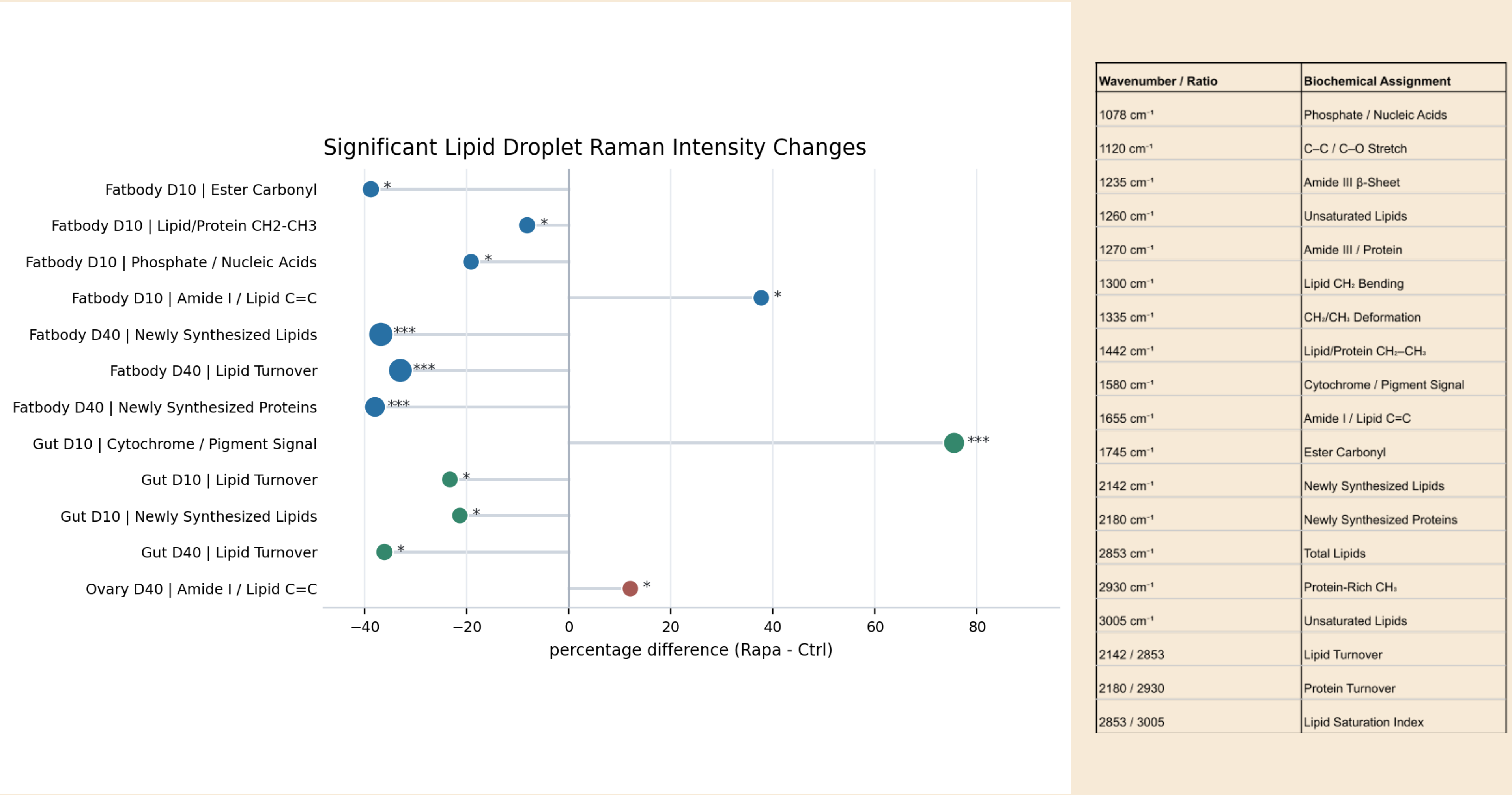


Figure 5. Lollipop plot of all significant peak intensity percent changes. All values (except for lipid and protein turnover) were normalized with intensity at 2930 cm⁻¹. Statistical significance was calculated from the Welch t statistic. A threshold of p < 0.05 was used to define statistical significance. Significance labels are shown as ns, p ≥ 0.05; *p < 0.05; **p < 0.01; ***p < 0.001; ****p < 0.0001.

Experimental Design

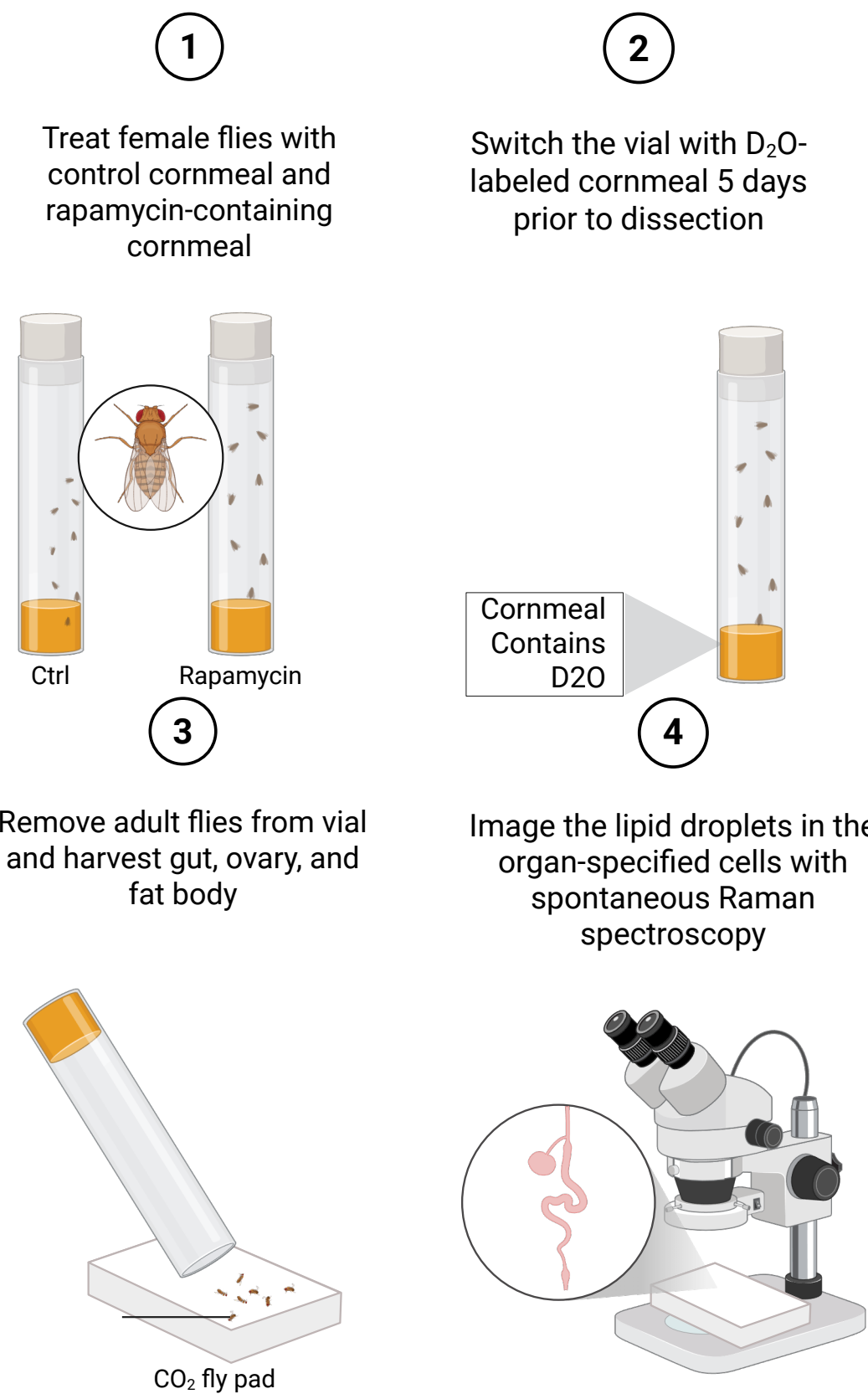


Figure 3. Experimental workflow for raising, dissecting, and imaging the Drosophila.

Conclusion

In D40 fat body, significant reduction in newly synthesized lipids, lipid turnover, and newly synthesized proteins with rapamycin suggests **suppression of biosynthetic activity in the fat body**. The gut also responded, particularly D10, with decreased lipid synthesis/turnover markers and a strong increase in a cytochrome/pigment-associated signal, indicating **early gut-specific biochemical remodeling**. In contrast, limited significant changes in ovary samples suggests either weaker sensitivity to rapamycin or greater sample heterogeneity. These results support the hypothesis that **rapamycin remodels lipid droplet composition and metabolic turnover in a tissue- and time-dependent manner**.

Future Direction

Spontaneous Raman Spectrometry is not a 2D mapping of the tissue, thus we might lose some spatial information. In the future, we are planning to use Stimulated Raman Scattering (SRS), for a better spatial resolution.

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